

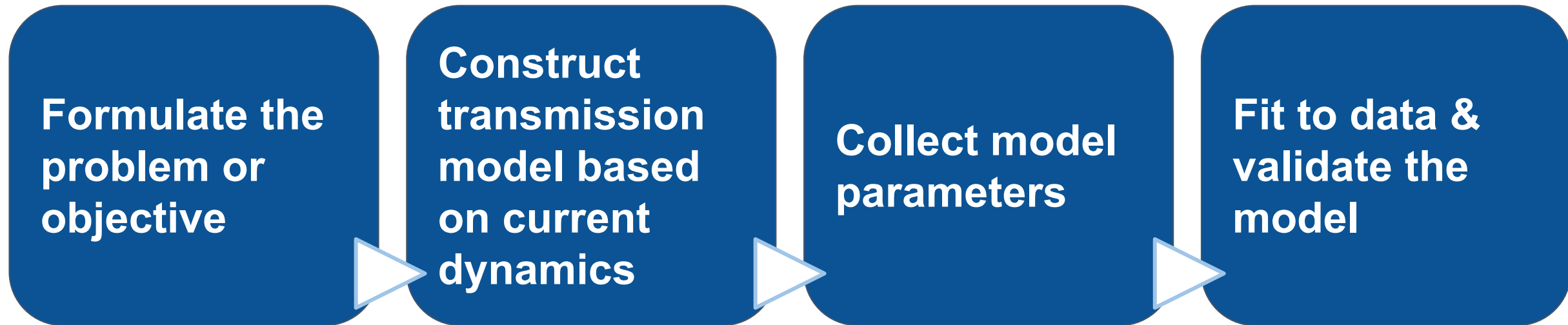
Stochastic Models

Recap

Days 1 & 2

RECAP

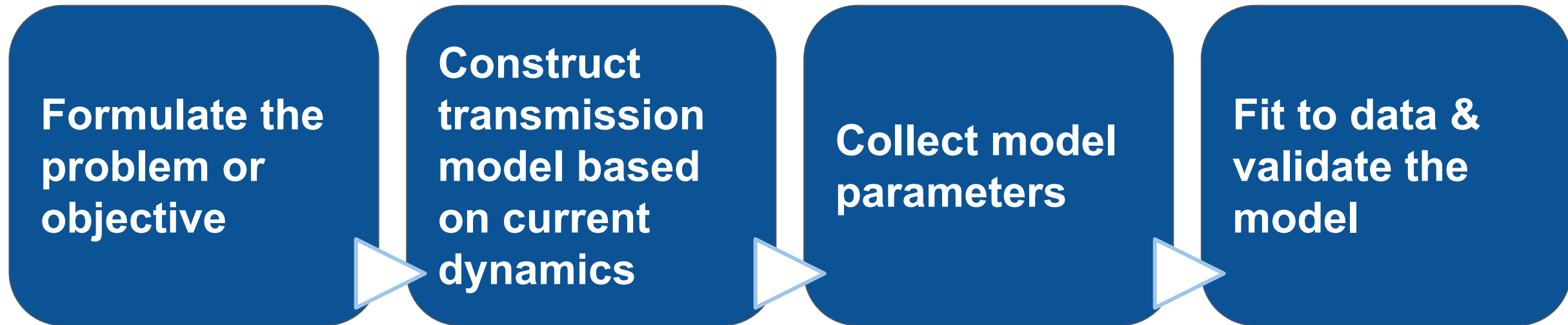
Steps of Developing a Model



- Objective informs model utility (*theoretical, inference, strategic, forecast*) informs model type

RECAP

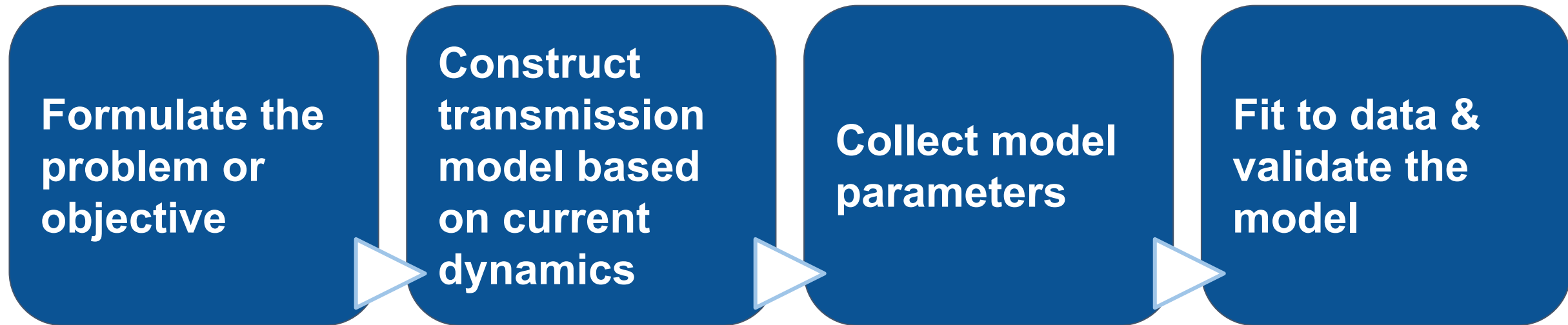
Steps of Developing a Model



- The basic SIR model
- Extension of the basic model:
vaccination activities
- **Modeling heterogeneity: *age-structured model***

RECAP

Steps of Developing a Model

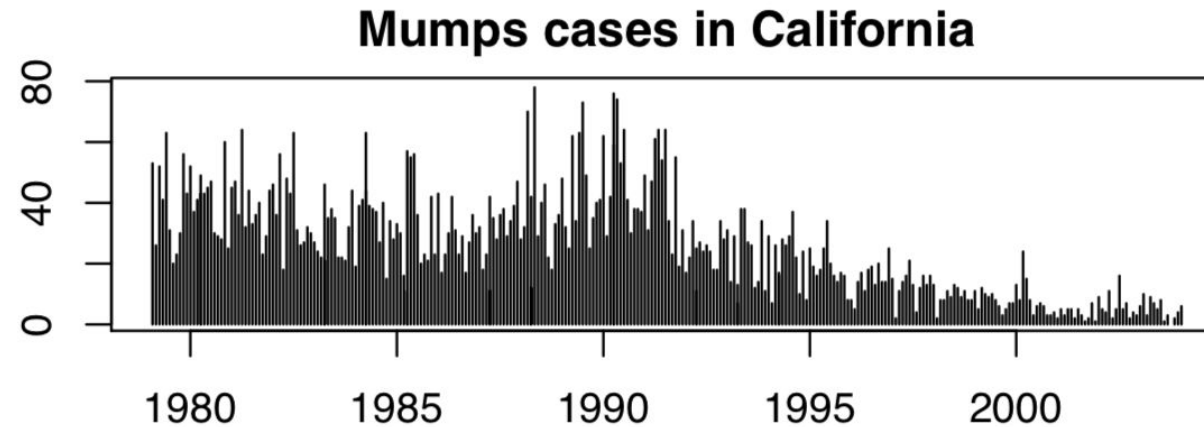


- How data are related to models
incidence, prevalence, seroprevalence
- Critical model parameters:
 R_0 and its components, β and γ
- **Estimating A and R_0**

Epidemiological Data are Noisy

Two types of noise:

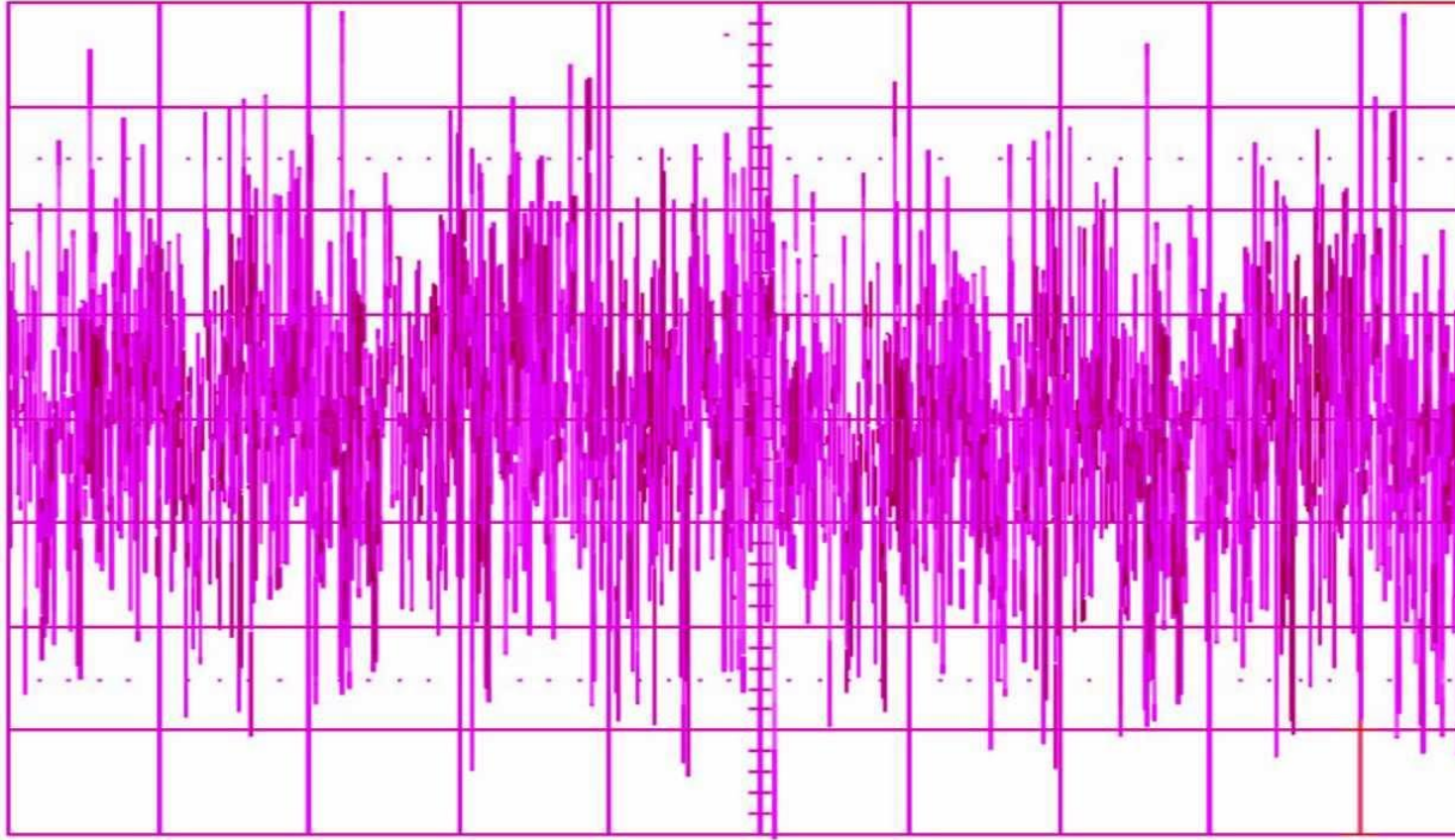
- Observation error: the data are probabilistically related to the true state of the system
- Process noise: the system progresses probabilistically
 - Environmental noise: some parameter is a random variable
 - Demographic noise: individual-level chance events



Noise is addressed using [stochastic models](#)

Deterministic Models

Deterministic models run like “clockwork”, given the same starting conditions, exactly-the-same trajectory will always be observed



Stochastic Models

Stochastic models aim to capture some of the random and probabilistic features of the real-world.

Stochastic models use random number generators, for example:

- parameter values pulled randomly from distributions
- random time-step length for events to occur
- reported cases pulled randomly from distributions with the mean being I_t^* (*mean report rate*)
- additional events not captured in the model structure (immigration, novel emergence)

Stochastic Simulation Algorithms

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Recall that:

Rate of event is proportional to time until event

Here, rates depend on current states (number of S/I/R)

So, rates are constant ***until*** the states change

When the states change, the rates change

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Recall that:

For constant rate processes, with rate θ , the time until the event occurs can be reasonably modeled as an exponentially distributed random variable with mean $1/\theta$

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

For this simple SIR model:

Two kinds of events can happen

1. An infection can occur, which leads to

$S \rightarrow S - 1$ and

$I \rightarrow I + 1$

This occurs at a rate βSI

2. An infected individual recovers, which leads to

$I \rightarrow I - 1$ and

$R \rightarrow R + 1$

This occurs at a rate γI

The Gillespie Algorithm

Conditional on current numbers of S, I, R and current time T

1. Take a random draws from Exponential distribution for all possible state transitions

X = random draw from exponential distribution with mean $1/\beta SI$

Y = random draw from exponential distribution with mean $1/\gamma I$

2. Update States: IF

$X \leq Y$, then transmission occurs first: $S \rightarrow S - 1$, $I \rightarrow I + 1$, increment time $T + X$

ELSE, recovery occurs first: $I \rightarrow I - 1$, $R \rightarrow R + 1$, increment time $T + Y$

3. Return to step 1

Interactive session 1: Run code

Questions:

What was R_0 ?

What did you notice about trajectories?

Gillespie Algorithm

Benefits

- Exact translation of stochastic ODE

Costs

- Computation scales with the number of state transitions (arrows in the SIR model) AND with the population size
- For this example, must draw ~ 2 random variables for every individual in the simulated population

Tau-leaping Algorithm

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Recall that:

IF an event occurs at a constant rate θ , THEN the number of events that will occur in a time interval ΔT will be distributed as:

Poisson($\theta * \Delta T$)

i.e. number of events will increase with the rate AND the duration of the interval

Tau-leaping Algorithm

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

So, for relatively small time steps, ΔT :

We can assume that the number of new infections that will occur will be distributed as:

$$\text{Poisson}(\beta SI * \Delta T)$$

And the number of recoveries that will occur will be distributed as:

$$\text{Poisson}(\gamma I * \Delta T)$$

Tau-leaping Algorithm

Conditional on current numbers of S, I, R and time step ΔT

1. Make random draws:

New infections, dSI , are random draw from $\text{Poisson}(\beta SI * \Delta T)$

New recoveries, dIR , are random draw from $\text{Poisson}(\gamma I * \Delta T)$

2. Update states

$$S_{\text{new}} = S - dSI$$

$$I_{\text{new}} = I + dSI - dIR$$

$$R_{\text{new}} = R + dIR$$

Update time by ΔT

3. Return to 1

Tau-leaping Algorithm

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Update time by ΔT

3. Return to 1

Recall the definitions of **Incidence** and **Prevalence**,

Which of the terms at the left is the **Incidence**?

Which of the terms at the left is the **Prevalence**?

Is the number of children newly diagnosed for measles at clinic **Incidence** or **Prevalence**?

Tau-leaping Algorithm

Benefits

- Computation (number of random draws) is independent of population size
(Note: while this makes a small difference in this code, it makes a **large** difference when simulating with births and deaths)

Costs

- Inexact. Requires assumption that rate is constant over ΔT , even though we know states are changing.

Interactive session 2: Run code

Questions:

How well do projections of Gillespie and Tau-leaping match?

Tau-leaping Algorithm

Benefits

- Computation (number of random draws) is independent of population size
(Note: while this makes a small difference in this code, it makes a **large** difference when simulating with births and deaths)

Costs

- Did anyone get “warning” errors?

There were 50 or more warnings (use warnings() to see the first 50)

Tau-leaping Algorithm

Conditional on current numbers of S, I, R and time step ΔT

1. Make random draws:

New infections, dSI , are random draw from $\text{Poisson}(\beta SI * \Delta T)$

New recoveries, dIR , are random draw from $\text{Poisson}(\gamma I * \Delta T)$

2. Update states

$$S_{\text{new}} = S - dSI$$

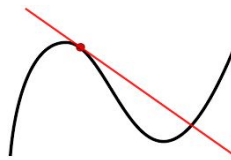
$$I_{\text{new}} = I + dSI - dIR$$

$$R_{\text{new}} = R + dIR$$

Update time by ΔT

3. Return to 1

Tangent



It is possible to make a random draw of dIR that is greater than I , which makes $I_{\text{new}} < 0$

This can be addressed by disallowing negative outcomes in code, or using binomial draws

Tau-leaping Algorithm

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

ALSO, for relatively small time steps, ΔT :

Rate in Poisson process can be translated to probability of event and modeled as a Binomial random variable.

The number of new infections that will occur will be:

$$\text{Binomial}(S, 1 - \exp(-\beta I * \Delta T))$$

And the number of recoveries that will occur will be:

$$\text{Binomial}(I, 1 - \exp(-\gamma * \Delta T))$$

Tau-leaping Algorithm

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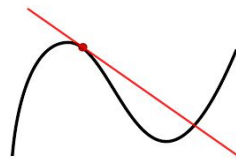
$$\text{Binomial}(I, 1 - \exp(-\gamma * \Delta T))$$

Recall that this expression turns **rate** (βSI or γ) measured over time into **risk**, which is measured per population.

ODEs, Gillespie, Tau-leaping

- An implicit assumption of the standard ODE representation the SIR model, the difference equation representation, AND both the Gillespie and Tau-leaping algorithms is that the rate of recovery (γ) is constant across time for each infected individual.
- This means that an individual is just as likely to recover after 1 day as after 10 days or 20 day (if they haven't already recovered)

Tangent



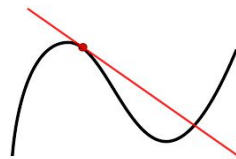
Some elegant approaches to overcome this:
http://alun.math.ncsu.edu/static/alun/tpb_01.pdf

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- This means that an individual is just as likely to recover after 1 day as after 10 days or 20 day (if they haven't already recovered)

One simple approach to overcome this is to use a time step, ΔT , that is equal to the average infectious period, $1/\gamma$

Tangent



Some elegant approaches to overcome this:
http://alun.math.ncsu.edu/static/alun/tpb_01.pdf

Chain Binomial Algorithm

If each infectious individual is expected to be infectious, for on average, a time equal to $1/\gamma$, then we can use a time step ΔT that is equal to $1/\gamma$.

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Why? ... well, this means that everyone is infected for the same amount of time (Gillespie and Tau-leaping technically allow some to be infected for very short or very long times.)

Chain Binomial Algorithm

If each infectious individual is expected to be infectious, for on average, a time equal to $1/\gamma$, then we can use a time step ΔT that is equal to $1/\gamma$.

Why? ... well, this means that everyone is infected for the same amount of time (Gillespie and Tau-leaping technically allow some to be infected for very short or very long times.)

And ... each case recovers at the end of each time step, so the number of Is in time step t is ***BOTH*** the prevalence ***AND*** the incidence. So this more naturally matches to observed time series of incident cases.

Chain Binomial Algorithm

Conditional on current numbers of S, I, R and time step equal to $1/\gamma$

1. I_{new} is a binomial draw from all current susceptibles

$$I_{\text{new}} \sim \text{binomial}(S, 1 - \exp(-\beta \frac{I}{N}))$$

2. $S_{\text{new}} = S - I_{\text{new}}$

3. Return to 1

Note that magnitude of β will be different than for Gillespie and Tau-leaping algorithms.

Expected value of binomial(N,p) is $N \cdot p$.

Here, the expected value of I_{new} is when $I=1$ and $S=N$ is

$$E[I_{\text{new}}] = S * 1 - \exp(-\beta \frac{I}{N}) \cong \beta * S * \frac{I}{N}$$

$$= N * 1 - \exp(-\beta \frac{1}{N}) = N * 1 - \exp(-\frac{\beta}{N})$$

$$\cong N * \frac{\beta}{N}$$

$$\cong \beta \rightarrow \beta = R_0$$

Chain Binomial Algorithm

Conditional on current numbers of S, I, R and time step equal to $1/\gamma$

1. I_{new} is a binomial draw from all current susceptibles

$$I_{\text{new}} \sim \text{binomial}(S, 1 - \exp(-\beta \frac{I}{N}))$$

2. $S_{\text{new}} = S - I_{\text{new}}$

3. Return to 1

Note that magnitude of β will be different than for Gillespie and Tau-leaping algorithms.

Expected value of $\text{binomial}(N, p)$ is $N \cdot p$.

Here, the expected value of I_{new} when $I=1$ and $S=N$ is

$$\begin{aligned} E[I_{\text{new}}] &= S * 1 - \exp(-\beta \frac{I}{N}) \cong \beta * S * \frac{I}{N} \\ &= N * 1 - \exp(-\beta \frac{1}{N}) = N * 1 - \exp(-\frac{\beta}{N}) \\ &\cong N * \frac{\beta}{N} \\ &\cong \beta \rightarrow \beta = R_0 \end{aligned}$$

Chain Binomial Algorithm

Benefits

- Computation (number of random draws) is independent of population size
- Only need 1 random draw per time step
- Natural translation to methods for inference ...

Costs

- Inexact. Requires assumption that rate is constant over ΔT , even though we know states are changing.

Interactive session 3: Run code

Questions:

How well do projections of Chain Binomial match Gillespie and Tau-leaping?

Chain Binomial Algorithm

Benefits

- Computation (number of random draws) is independent of population size
- Randomness is limited only to the infection
- Individuals all remain infectious for the same amount time

Costs

- Inexact. Requires assumption that rates are constant over the entire epidemic

Chain Binomial Algorithm

Conditional on current numbers of S, I, R and time step equal to $1/\gamma$

1. I_{new} is a binomial draw from all current susceptibles

$$I_{\text{new}} \sim \text{binomial}(S, 1 - \exp(-\beta \frac{I}{N}))$$

2. $S_{\text{new}} = S - I_{\text{new}}$

3. Return to 1

Note that magnitude of β will be different than for Gillespie and Tau-leaping algorithms.

Expected value of binomial(N,p) is $N \cdot p$.

Here, the expected value of I_{new} is when $I=1$ and $S=N$ is

$$E[I_{\text{new}}] = S * 1 - \exp(-\beta \frac{I}{N}) \cong \frac{\beta}{N} * S * I = \beta * S * I$$

$$= N * 1 - \exp(-\beta \frac{1}{N}) = N * 1 - \exp(-\frac{\beta}{N})$$

β is $\sim R_0$

$$\cong \beta \rightarrow \beta = R_0$$

Chain Binomial Algorithm – with births

Conditional on current numbers of S, I, R and time step equal to $1/\gamma$

1. I_{new} is a binomial draw from all current susceptibles

$$I_{\text{new}} \sim \text{binomial}(S, 1 - \exp(-\beta \frac{I}{N}))$$

2. $S_{\text{new}} = B + S - I_{\text{new}}$

Note that magnitude of β will be different than for Gillespie and Tau-leaping algorithms.

3. Return to 1

Expected value of binomial(N,p) is $N \cdot p$.

Here, the expected value of I_{new} is when $I=1$ and $S=N$ is

$$E[I_{\text{new}}] = S * 1 - \exp(-\beta \frac{I}{N}) \cong \frac{\beta}{N} * S * I = \beta * S * I$$

B = birth in each time step

$$\begin{aligned} &= N * 1 - \exp(-\beta \frac{1}{N}) = N * 1 - \exp(-\frac{\beta}{N}) \\ &\cong N * \frac{\beta}{N} \\ &\cong \beta \rightarrow \beta = R_0 \end{aligned}$$

Break to old slides